

生态研究前沿速览_2026-03-12

【土壤与生物地球化学】

1. Tropical wet season runoff mobilises younger carbon in rainforest streams but older carbon in agricultural streams

□ Biogeosciences | DOI: <https://doi.org/10.5194/bg-23-1755-2026>

□ 摘要: Tropical wet season runoff mobilises younger carbon in rainforest streams but older carbon in agricultural streams Clément Duvert, Vanessa Solano, Dioni I. Cendón, Francesco Ulloa-Cedamano, Liza K. McDonough, Robert G. M. Spencer, Niels C. Munksgaard, Lindsay B. Hutley, Jean-Sébastien Moquet, and David E. Butman Biogeosciences, 23, 1755–1770, <https://doi.org/10.5194/bg-23-1755-2026>, 2026 This study examines the age and composition of carbon in tropical streams. We find that dissolved organic ca...

2. Detection of dietary stress and geophagic behaviour forced by dry seasons in Miocene Gomphotherium

□ Biogeosciences | DOI: <https://doi.org/10.5194/bg-23-1833-2026>

□ 摘要: Detection of dietary stress and geophagic behaviour forced by dry seasons in Miocene Gomphotherium Rute Coimbra, Niels de Winter, Maria Ríos, Rui Bernardino, Darío Estraviz-López, Priscila Lohmann, Roberta Martino, Aurora Grandal-d'Anglade, Fernando Rocha, and Philippe Claeys Biogeosciences, 23, 1833 – 1858, <https://doi.org/10.5194/bg-23-1833-2026>, 2026 To understand human impact on climate and biodiversity, we studied fossil teeth of Gomphotherium from Miocene Portugal. Chemical patterns, like th...

3. On the role of light and vertical mixing in shaping Southwestern Atlantic shelf blooms

□ Biogeosciences | DOI: <https://doi.org/10.5194/bg-23-1859-2026>

□ 摘要: On the role of light and vertical mixing in shaping Southwestern Atlantic shelf blooms Ana I. Dogliotti, Reinaldo A. Maenza, Moira Luz Clara, Vivian A. Lutz, and Robert Frouin Biogeosciences, 23, 1859 – 1880, <https://doi.org/10.5194/bg-23-1859-2026>, 2026 We analyzed 22 years of satellite and modeled data to study how light and mixing shape phytoplankton blooms on the Argentine Continental Shelf. Blooms start earlier on the central shelf and coast, and later on the deeper, colder Patagonian Shelf. ...

4. Straw Incorporation Strategies Regulate Detritosphere Organic Carbon and Nutrient Accumulation

□ European Journal of Soil Science | DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70297>

□ 摘要: European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

5. Evaluating glycerol dialkyl glycerol tetraether (GDGT)-based reconstructions from varved lake sediments during the Holocene

□ Biogeosciences | DOI: <https://doi.org/10.5194/bg-23-1809-2026>

□ 摘要: Evaluating glycerol dialkyl glycerol tetraether (GDGT)-based reconstructions from varved lake sediments during the Holocene Ashley M. Abrook, Gordon N. Inglis, Peter G. Langdon, McKenzie R. Bentley, Achim Brauer,

Ian Bull, Daisy Fallows, Paul Lincoln, Antti E. K. Ojala, Helen L. Whelton, and Celia Martin-Puertas Biogeosciences, 23, 1809–1832, <https://doi.org/10.5194/bg-23-1809-2026>, 2026 We present lipid-based environmental and temperature reconstructions spanning the Holocene from three Europea...

6. Soil Carbon Modeling at Crossroads: Building Reliable Methods for Policy and Practice

☐ European Journal of Soil Science | DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70295>

☐ 摘要: European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

7. Species-specific differential dissolution morphology of selected coccolithophore species: an experimental study

☐ Biogeosciences | DOI: <https://doi.org/10.5194/bg-23-1795-2026>

☐ 摘要: Species-specific differential dissolution morphology of selected coccolithophore species: an experimental study Gerald Langer, Ian Probert, Jeremy R. Young, and Patrizia Ziveri Biogeosciences, 23, 1795 – 1808, <https://doi.org/10.5194/bg-23-1795-2026>, 2026 Coccolithophores are important marine CaCO₃ producers and their biominerals, the coccoliths, partly dissolve in the upper water column where dissolution is unexpected. Studying coccolith dissolution in field samples is hampered by a paucity of ex...

8. Soil Management Effects on Grapevine Water Uptake Depth in a Mediterranean Vineyard

☐ European Journal of Soil Science | DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70301>

☐ 摘要: European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

9. Subduing the Deluge in Soil Science: What If Each Researcher Published Only One Paper Per Year?

☐ European Journal of Soil Science | DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70296>

☐ 摘要: European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

10. Contrasting Effects of Long-Term Fertilization on Pore Structure and POM in Upland and Paddy Soils Revealed by X-Ray CT

☐ European Journal of Soil Science | DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70298>

☐ 摘要: European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

11. Nitrate sources and transformation pathways in salt-affected agricultural ditches: biogeochemical responses along contrasting salinity gradients

☐ Biogeochemistry | DOI: <https://link.springer.com/article/10.1007/s10533-026-01318-y>

☐ 摘要: Agricultural nitrate pollution, largely associated with high-intensity fertilizer application and untreated or insufficiently treated waste inputs, has become a major driver of water quality degradation worldwide. Agricultural ditches, as key recipients of surface runoff and lateral seepage, function as both transport conduits and biogeochemical hotspots for nitrogen transformation. In salt-affected ditches, high nitrogen inputs and salinity can interact to enhance nitrate accumulation and mobil...

12. No indications for a priming effect on soil organic carbon mineralization in a temperate river system

☐ Biogeochemistry | DOI: <https://link.springer.com/article/10.1007/s10533-026-01317-z>

☐ 摘要: Globally, rivers are vital conduits transporting and processing terrestrial carbon, and are generally considered to act as source of carbon dioxide (CO₂) towards the atmosphere. A large amount of soil organic carbon

(SOC) is transferred from the land surface to river systems each year, where it mixes and interacts with the autochthonous carbon pool (i.e., produced in-stream via photosynthesis). The latter has been suggested to be more labile and to potentially affect—positively or negatively—the...

13. Precipitation legacy effects on gross N mineralization and nitrification rates in a Pinyon-Juniper dryland under altered precipitation

☐ Biogeochemistry | DOI: <https://link.springer.com/article/10.1007/s10533-026-01313-3>

☐ 摘要: Changes in precipitation patterns can influence soil nitrogen (N) cycling by altering the timing and extent of moisture availability. Arid conditions, associated with more frequent and severe droughts, can increase soil inorganic N concentrations when diffusion constraints decouple N sources and sinks, making this N susceptible to loss. Further, shifts in precipitation patterns in one season can “carry over” N into subsequent seasons, known as precipitation legacies. However, how these legacies ...

14. Soil heterotrophic respiration after irrigation retirement is differentially influenced by moisture and substrate availability over time

☐ Biogeochemistry | DOI: <https://link.springer.com/article/10.1007/s10533-026-01312-4>

☐ 摘要: Water limitations are forcing producers to transition large areas of currently irrigated farmland into dryland agriculture across the Western U.S. with unclear effects on global soil carbon (C) dynamics. An experiment established in 2017 in a no-till, maize system in Colorado suggested that soil heterotrophic respiration (Rh) following irrigation retirement was co-regulated by water and available C. We continued Rh measurements in 2021–2022 along with monthly soil samplings to explore the intera...

15. Mechanisms and sources of organic carbon accrual in deep soil under long-term straw return

☐ Soil & Tillage Research | DOI: <https://www.sciencedirect.com/science/article/pii/S0167198726001066>

☐ 摘要: Publication date: September 2026Source: Soil and Tillage Research, Volume 261Author(s): Wenhao Feng, Jie Zhou, Antonio Rafael Sánchez-Rodríguez, Yakov Kuzyakov, Ji Chen, Shang Wang, Haoruo Li, Jing Wang, Hongyuan Zhang, Roland Bol, Yuyi Li, Leanne Peixoto

16. Long-term N fertilization altered viral community composition and increased functional potentials of phosphorus cycling by causing soil acidification

☐ Soil Biology and Biochemistry | DOI: <https://www.sciencedirect.com/science/article/pii/S0038071726000568>

☐ 摘要: Publication date: June 2026Source: Soil Biology and Biochemistry, Volume 217Author(s): Shuxun Cheng, Zhongmin Dai, Dan He, Ran Xue, Kankan Zhao, Jorge L. Mazza Rodrigues, Lucas W. Mendes, Scott X. Chang, Jianming Xu

【预印本】

1. Climate change intensifies rapid genomic selection beyond the ancestral niche of *Fagus sylvatica*

☐ bioRxiv Evolutionary Biology | DOI: <https://doi.org/10.1101/2026.03.09.710448>

☐ 摘要: As climate change accelerates, the persistence of long-lived organisms increasingly depends on their capacity to adapt in situ. While phenotypic plasticity provides an immediate buffer, it remains uncertain whether forest trees can evolve rapidly enough to track shifting climatic niches. Here, we investigate the adaptive potential of European beech (*Fagus sylvatica* L.), a keystone temperate species, by leveraging different growth classes as a quasi-time-series. This approach allows us to compare...

2. Combined impacts of invasive alien species and fire on ecosystems are complex, mostly negative, and understudied: a global review

☐ bioRxiv Ecology | DOI: <https://doi.org/10.1101/2026.03.08.710355>

☐ 摘要: Invasion-fire interactions can negatively impact ecosystems by driving biodiversity loss, altering ecological processes, modifying habitat structure, and compromising ecosystem functioning. Understanding how this interaction operates is essential to design effective management strategies that are successful in controlling both invasive alien species (IAS) and fire. Therefore, the present literature review aims to synthesize the current knowledge on invasion-fire interaction and its impacts on ec...

3. First draft genome of the decaploid species, *Ludwigia grandiflora* subsp. *hexapetala*, validated through gene expression

☐ bioRxiv Genomics | DOI: <https://doi.org/10.1101/2026.03.09.710498>

☐ 摘要: Invasive species are one of the biggest drivers of species extinction. *Ludwigia grandiflora* subsp. *hexapetala* (Lgh) is widely invasive in aquatic ecosystems of Europe, North America, and Japan, and also colonizes emergent freshwater soils, but limited genomic data constrain studies of its invasiveness. Here, we report a draft genome assembly of Lgh, with a total length of 1.487 Gb, in agreement with the genome size estimated by flow cytometry, despite high fragmentation (111,219 contigs; N50 = 1...

4. Predicting life-history traits in a stored bean pest beetle *Callosobruchus chinensis* (Coleoptera: Chrysomelidae: Bruchinae) using machine learning

☐ bioRxiv Ecology | DOI: <https://doi.org/10.1101/2026.03.07.710260>

☐ 摘要: Life-history traits play an important role in insect population dynamics and ecological processes. The azuki bean beetle *Callosobruchus chinensis* is a common pest of stored legumes and is also widely used as a model species in ecological and evolutionary research. In this study, we tested whether machine learning models could be used to estimate several traits of *C. chinensis*, including elytral length, development time and adult lifespan. Experimental data were obtained from laboratory population...

5. Implications of the trade-offs between negative density-dependence and Allee effects for vector control

☐ bioRxiv Ecology | DOI: <https://doi.org/10.1101/2026.03.07.710288>

☐ 摘要: Understanding population dynamics is fundamental to predicting species persistence and extinction, yet remains challenging due to the complex interplay between ecological, environmental and anthropogenic factors. Population dynamics are regulated by intrinsic factors such as negative density-dependence and Allee effects. While negative density-dependence is a well-understood process, Allee effects have received less attention but can have important implications for conservation and species management...

6. Photodegradation accelerates standing dead litter decomposition in monsoonal mountain grasslands of South America

☐ bioRxiv Ecology | DOI: <https://doi.org/10.1101/2026.03.06.709891>

☐ 摘要: Plant litter decomposition is the process through which plant-derived organic matter is recycled in terrestrial ecosystems. One of the drivers of decomposition is photodegradation, light-induced reactions that result in litter mass loss and transformations that accelerate the decomposition process. Photodegradation has been mostly studied in arid ecosystems, but mesic grasslands with monsoonal climate have been almost absent in the literature. In these ecosystems, standing dead biomass might remain...

7. Cellular transcriptomics reveals evolutionary adaptation and rumination of vertebrate stomachs

☐ bioRxiv Evolutionary Biology | DOI: <https://doi.org/10.1101/2026.03.09.710455>

☐ 摘要: Stomach, an essential digestive organ in vertebrates, has undergone significant changes in its morphology, chamber numbers, and physiology, in response to dietary diversity. However, its cellular and molecular basis underlying evolutionary adaptation remains largely uncharacterized. Here we report a single cell and spatial transcriptomic atlas of stomachs from 23 vertebrate species with diverse feeding habits (e.g., omnivores, carnivores, and herbivores) and gastric architectures, including mono...

8. Characterizing Physicochemical Selection in Protein Evolution with Property-Informed Models (PRIME)

☐ bioRxiv Evolutionary Biology | DOI: <https://doi.org/10.1101/2026.03.09.710461>

☐ 摘要: Standard probabilistic models of coding sequence evolution effectively identify where and when selection acts but remain agnostic to the mechanistic realization of these forces. We introduce PRIME (PRoperty Informed Models of Evolution), a framework of codon-level maximum likelihood methods - including global (G-PRIME), episodic (E-PRIME), and site-specific (S-PRIME) implementations - that explicitly model amino acid exchangeability as a function of physicochemical properties. By parameterizing ...

9. TCA cycle entry point, growth variability and amino acid utilization in *Alteromonas macleodii* ATCC 27126

☐ bioRxiv Ecology | DOI: <https://doi.org/10.1101/2026.03.04.709670>

☐ 摘要: Amino acid catabolism is a vital metabolic process in bacteria, providing energy, carbon and potentially nitrogen as resources, and affecting global cycles of these elements. The ability of a bacterium to catabolize an amino acid is often inferred from the presence of the relevant catabolic pathways in its genome, yet the "gene=function" inference is not straightforward. Here, we use growth assays in 96 well plates on individual amino acids and their combinations to directly measure the ability ...

10. Uncovering Carbohydrate Metabolism and Endogenous Hormone Regulation during Flush Phenology in Citrus Trees using Proteomics and Metabolomics

☐ bioRxiv Plant Biology | DOI: <https://doi.org/10.1101/2026.03.09.710605>

☐ 摘要: Rapid shoot growth (flushing) phenology is a fundamental developmental process in perennial woody plants such as citrus. In a separate study, we identified physiological shifts from photosynthesis to mobilization of nitrogen and carbohydrate to support new shoot growth. However, the underlying molecular and biochemical signals remain largely unknown. Here, we integrated proteomic and metabolomic analyses to investigate carbohydrate and hormone dynamics across three flush stages in *Citrus sinensis*...

11. Is It Time for Medical Social Work Training to Evolve? Insight Gained during COVID-19

☐ Research Square - Environment | DOI: <https://www.researchsquare.com/article/rs-8759118/latest>

☐ 摘要: Background: The COVID-19 pandemic exposed significant gaps in the preparation of medical social workers (MedSWs) for practice in complex healthcare environments. Despite MedSW being one of the largest and fastest-growing areas of social work practice, training remains inconsistent and is often relegated to elective coursework. This study examines how MedSW curricula can evolve to better prepare practitioners for post-pandemic healthcare systems. Methods: This qualitative study drew on semi-struct...

12. Unsupervised explainable AI reveals similar oligonucleotide-usage zones matching the highest-resolution human chromosome bands

☐ bioRxiv Genomics | DOI: <https://doi.org/10.1101/2026.03.09.710446>

☐ 摘要: Unsupervised and explainable AI can uncover genomic features that extend beyond human expectations. Oligonucleotides such as penta- and hexanucleotides often function as core binding motifs for regulatory proteins, and their usage provides a powerful tool in functional genomics. We applied an unsupervised, explainable AI

approach to odds ratio (observed/expected) profiles of all 1-Mb euchromatic fragments in the human genome. This odds-ratio analysis identified oligonucleotide features indepen...

13. A robust and high-efficiency *Rhizobium rhizogenes* hairy root transformation platform for *Vaccinium*

☐ bioRxiv Plant Biology | DOI: <https://doi.org/10.1101/2026.03.09.710070>

☐ 摘要: Efficient transformation remains a major constraint to functional genomics and genome editing in *Vaccinium*, where stable transformation systems are often genotype-dependent and inefficient. Here, we establish a rapid and high-efficiency *Rhizobium rhizogenes*-mediated hairy root transformation platform optimized for the genus. Using the RUBY visual reporter, transformation efficiency reached 46.7% in leaf explants infected with strain Ar. A4 and cultured on half-strength Woody Plant Medium, with t...

【生态学核心】

1. Global Marine Fishery Stock Productivity Under Climate Change

☐ Global Change Biology | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70784>

2. Heat Tolerance and Its Plasticity in Freshwater and Marine Fishes Reflect Exposure to Extremes and Seasonal Variation in Habitat Temperatures

☐ Ecology Letters | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70341>

3. Influences of Structural and Species Diversity on Forest Resistance to Drought

☐ Ecology Letters | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70351>

4. Drained Agricultural Peatlands as Persistent Carbon Sources: Implications for Carbon and Water Use Intensity in Food Production

☐ Global Change Biology | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70796>

5. Genetic ancestry affects cancer immunity

☐ Nature Ecology & Evolution | DOI: <https://doi.org/10.1038/s41559-026-03012-6>

☐ 摘要: Nature Ecology & Evolution, Published online: 11 March 2026; doi:10.1038/s41559-026-03012-6 Genetic ancestry affects cancer immunity

6. Forest Heterogeneity by Chain Saw: How Between-Patch Variation in Old Growth Attributes Changes the Metacommunities of Beetles

☐ Ecology Letters | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70355>

7. A Global Conservation Blind Spot: Neglect of Bryophytes Undermines Biodiversity Targets

☐ Global Change Biology | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70789>

8. Plasticity in prostate cancer

☐ Nature Ecology & Evolution | DOI: <https://doi.org/10.1038/s41559-026-03024-2>

☐ 摘要: Nature Ecology & Evolution, Published online: 11 March 2026; doi:10.1038/s41559-026-03024-2Plasticity in prostate cancer

9. Drift shapes norovirus diversity

☐ Nature Ecology & Evolution | DOI: <https://doi.org/10.1038/s41559-026-03022-4>

☐ 摘要: Nature Ecology & Evolution, Published online: 11 March 2026; doi:10.1038/s41559-026-03022-4Drift shapes norovirus diversity

10. Aneuploidy affects fitness

☐ Nature Ecology & Evolution | DOI: <https://doi.org/10.1038/s41559-026-03016-2>

☐ 摘要: Nature Ecology & Evolution, Published online: 11 March 2026; doi:10.1038/s41559-026-03016-2Aneuploidy affects fitness

11. Evolution in medicine

☐ Nature Ecology & Evolution | DOI: <https://doi.org/10.1038/s41559-026-03028-y>

☐ 摘要: Nature Ecology & Evolution, Published online: 11 March 2026; doi:10.1038/s41559-026-03028-yIn this Focus issue, we highlight research and opinion pieces on the many conceptual and practical contributions of evolutionary biology to predicting and controlling disease.

12. Anaemic Streams: Iron and Essential Trace Metals Frequently Limit Primary Producer Biomass

☐ Ecology Letters | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70357>

13. There is No Consensus on Biological Sex

☐ Ecology Letters | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70350>

14. Retrogressive Thaw Slumps Produce a Changing Disturbance Regime for Arctic Stream Invertebrates

☐ Global Change Biology | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70701>

15. Widespread Shrubification on European Mountain Summits

☐ Global Change Biology | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70786>

16. A Bias-Robust Framework for Quantifying Community Responses to the Climate Change Using the Occurrence Data

☐ Global Ecology and Biogeography | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/geb.70223>

17. Density dependence modulates climate change impacts on eastern Bering Sea snow crab

☐ Journal of Applied Ecology | DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2664.70316>

18. Geodiversity is an inseparable but underutilized aspect of ecological connectivity assessments under climate change

☐ Ecography | DOI: <https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/ecog.07768>

☐ 摘要: Ecography, EarlyView.

19. Genomic life-history traits mediate the impacts of habitat loss and nutrient availability on fragmented grassland communities

☐ Journal of Ecology | DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70291>

20. As above, so below: A perspective into the application of land-forest monitoring methods for the assessment of marine animal forests

☐ Methods in Ecology and Evolution | DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210x.70242>

☐ 摘要: Methods in Ecology and Evolution, EarlyView.

21. Extending plant water-use strategies to flowers: Evidence from trait correlations across plant organs

☐ Functional Ecology | DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.70293>

☐ 摘要: Functional Ecology, EarlyView.

22. The post-fire recovery of soil seed banks along a fire severity gradient in an Australian threatened mesic forest

☐ Functional Ecology | DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.70296>

☐ 摘要: Functional Ecology, EarlyView.

23. High temperatures and poor habitat reduce nestling condition and survival in a tropical songbird

☐ Journal of Applied Ecology | DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2664.70322>

24. Biochar enhances blue carbon stability in seagrass ecosystems

☐ Journal of Applied Ecology | DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2664.70320>

25. A transcontinental experiment elucidates (mal)adaptation of a cosmopolitan plant to climate in space and time

☐ Ecological Monographs | DOI: <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.70043>

26. Does restoring apex predators to food webs restore ecosystems? Reply

☐ Ecological Monographs | DOI: <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.70049>

27. Does restoring apex predators to food webs restore ecosystems? Comment

☐ Ecological Monographs | DOI: <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.70050>

28. Plant interactions, climate, and the reindeer (*Rangifer tarandus*) interdependently shape vegetation in northern Finland

☐ Ecological Applications | DOI: <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/eap.70200>

【环境与气候】

1. Changing bird nutrient inputs

☐ Nature Climate Change | DOI: <https://doi.org/10.1038/s41558-026-02592-2>

☐ 摘要: Nature Climate Change, Published online: 10 March 2026; doi:10.1038/s41558-026-02592-2 Changing bird nutrient inputs

2. Policy interactions reshape the outcomes of carbon pricing policies

☐ Nature Climate Change | DOI: <https://doi.org/10.1038/s41558-026-02578-0>

☐ 摘要: Nature Climate Change, Published online: 11 March 2026; doi:10.1038/s41558-026-02578-0 The adoption and effectiveness of carbon pricing are highly reshaped by interactions with other climate mitigation policies. A global comparative assessment of policy synergies and conflicts can guide policymakers in designing policy portfolios that can achieve higher mitigation cost-effectiveness.

3. Cross-national comparative assessment of synergies and conflicts in climate policy mixes

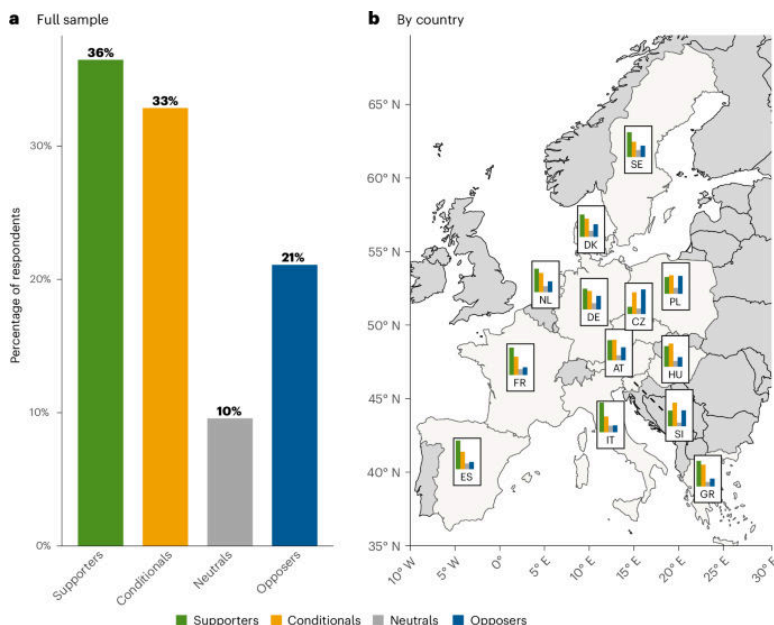
☐ Nature Climate Change | DOI: <https://doi.org/10.1038/s41558-026-02574-4>

☐ 摘要: Nature Climate Change, Published online: 11 March 2026; doi:10.1038/s41558-026-02574-4 Interactions between climate policy instruments can have synergistic and conflicting effects, but these interactions are not systematically understood. This research provides global evidence on how policy characteristics and interactions in different contexts could lead to different outcomes.

4. Climate policy feasibility across Europe relies on the conditional middle

☐ Nature Climate Change | DOI: <https://doi.org/10.1038/s41558-026-02562-8>

☐ 摘要: Nature Climate Change, Published online: 11 March 2026; doi:10.1038/s41558-026-02562-8 The feasibility of climate policies hinges on public support. A survey of 13 EU countries shows that ‘middle groups’—citizens whose support across mitigation measures varies, rather than being uniformly supportive or opposed—play a pivotal role in shaping overall public policy support and electoral outcomes.



5. Amplified variability

☐ Nature Climate Change | DOI: <https://doi.org/10.1038/s41558-026-02593-1>

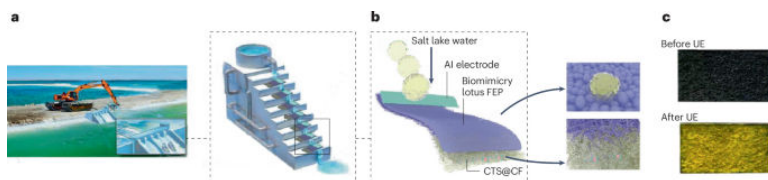
☐ 摘要: Nature Climate Change, Published online: 10 March 2026; doi:10.1038/s41558-026-02593-1 Amplified variability



6. Nuclear fuel from saline waters

☐ Nature Sustainability | DOI: <https://doi.org/10.1038/s41893-026-01790-2>

☐ 摘要: Nature Sustainability, Published online: 05 March 2026; doi:10.1038/s41893-026-01790-2 Growing demand for nuclear fuel is accelerating the need for sustainable uranium resources that ensure long-term supply and mitigate environmental risks. A study now presents an intriguing self-powered methodology that utilizes the motion of falling water droplets to drive uranium recovery from salt lakes.



7. Compostable soft robots for plant monitoring

☐ Nature Sustainability | DOI: <https://doi.org/10.1038/s41893-025-01757-9>

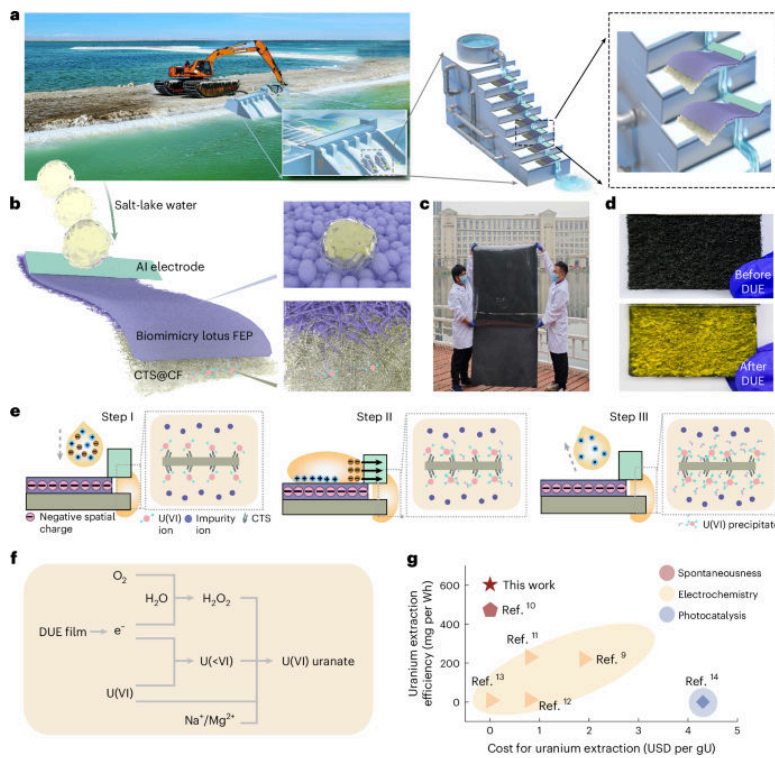
☐ 摘要: Nature Sustainability, Published online: 05 March 2026; doi:10.1038/s41893-025-01757-9 Soft robots inspired by living organisms hold the promise of gentle, adaptable interactions with the natural world, but leave behind persistent waste. Now scientists show a fully compostable robotic system that addresses this limitation by offering durable performance and decomposing safely into the soil at the end of its life.



8. Spontaneous uranium extraction from salt-lake water via droplet falling

□ Nature Sustainability | DOI: <https://doi.org/10.1038/s41893-026-01781-3>

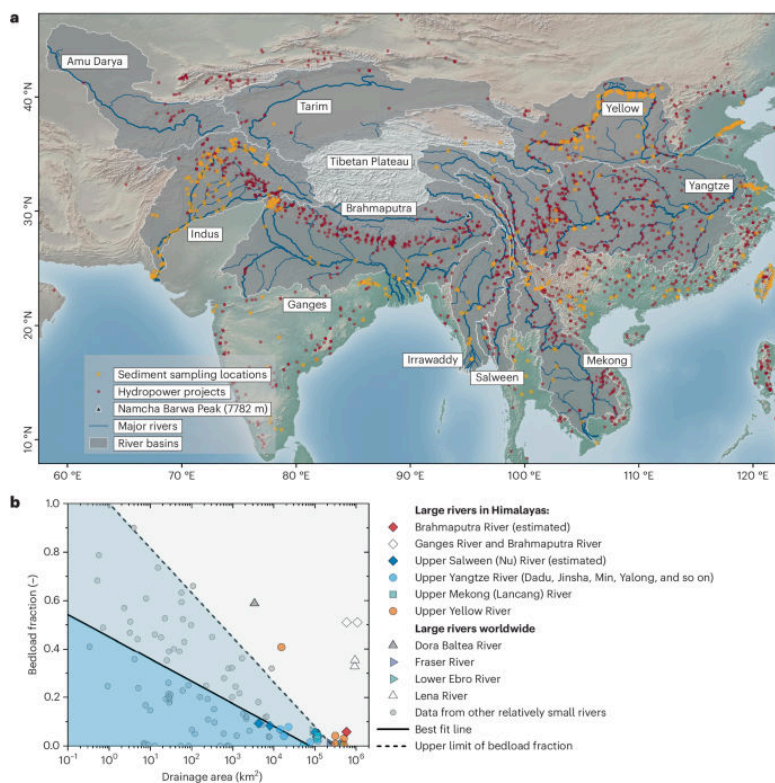
□ 摘要: Nature Sustainability, Published online: 05 March 2026; doi:10.1038/s41893-026-01781-3 More environmentally friendly methods of extracting uranium from alternative sources are needed for the sustainable development of the nuclear power industry. This work proposes a spontaneous extraction process driven by droplet falling that enables the extraction of uranium from salt lakes.



9. Overlooked bedload transport in Himalayan rivers threatens regional security

□ Nature Geoscience | DOI: <https://doi.org/10.1038/s41561-026-01927-0>

□ 摘要: Nature Geoscience, Published online: 02 March 2026; doi:10.1038/s41561-026-01927-0 Bedload transport in high Himalayan rivers has been overlooked, hindering the development of morphodynamic models that explicitly couple river hydraulics, sediment transport, and channel morphology to predict responses to extreme floods under global warming. This knowledge gap increases risks to regional hydropower, ecosystems, and food security.



10. Biodegradable yet hyperdurable robotic fingers for zero-waste soft electronics

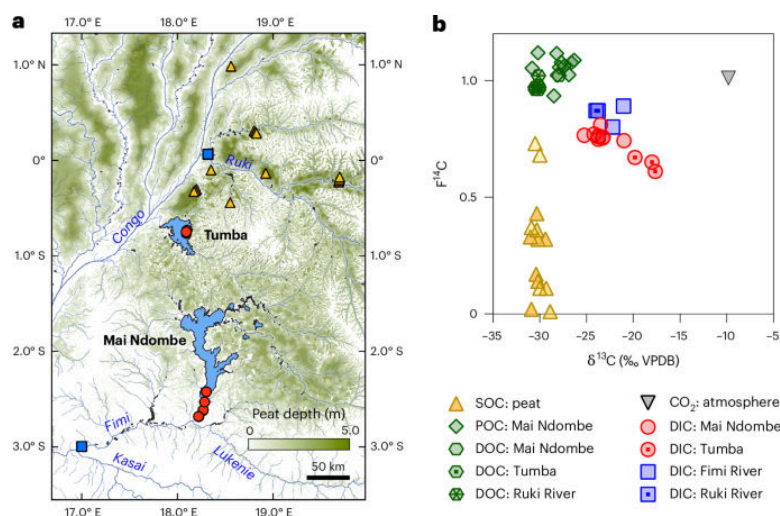
☐ Nature Sustainability | DOI: <https://doi.org/10.1038/s41893-026-01780-4>

☐ 摘要: Nature Sustainability, Published online: 05 March 2026; doi:10.1038/s41893-026-01780-4 As soft electronic waste becomes an urgent concern, biodegradable yet high-performance devices are emerging as a promising solution. Here the authors fabricate durable and multifunctional soft robotic fingers in which both polymers and inorganic electronics are fully compostable.

11. Congo's largest humic lakes emit ancient peat carbon

☐ Nature Geoscience | DOI: <https://doi.org/10.1038/s41561-026-01925-2>

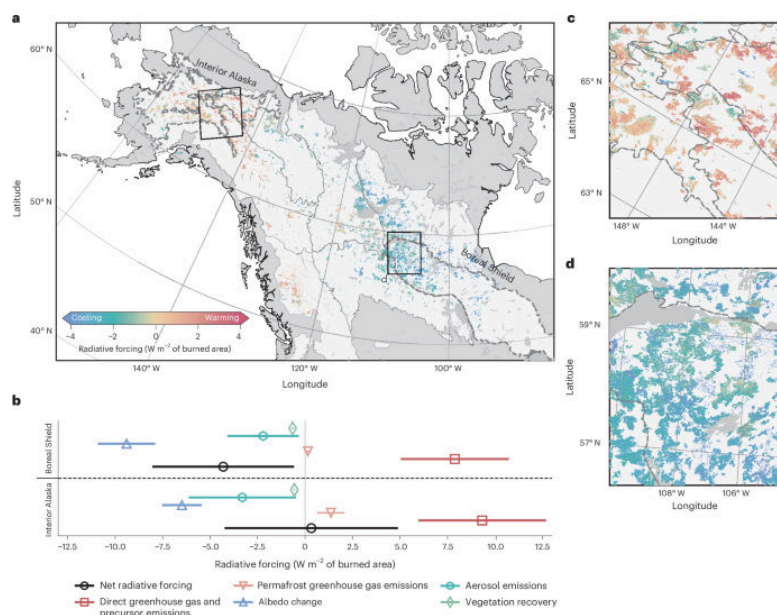
☐ 摘要: Nature Geoscience, Published online: 04 March 2026; doi:10.1038/s41561-026-01925-2 We revealed that carbon dioxide emissions from the Congo Basin's largest humic lakes partially originate from millennia-old peat. This discovery identifies a pathway whereby deep, long-sequestered carbon is respired and mobilized by subsurface flows, indicating these massive reservoirs of organic carbon are less inert than previously assumed.



12. Climate impacts from North American boreal forest fires

□ Nature Geoscience | DOI: <https://doi.org/10.1038/s41561-026-01940-3>

□ 摘要: Nature Geoscience, Published online: 03 March 2026; doi:10.1038/s41561-026-01940-3 While boreal forest fires in Alaska tend to induce net warming through high fuel consumption and permafrost thaw, those in Canada generally lead to net cooling by increasing snow albedo, according to a multi-data analysis over 2001–2019.



13. [ASAP] Coordination Microenvironment Engineering in Nanoconfined Structures Shifts Peracetic Acid Activation toward Broad-Spectrum Radicals for Enhanced Hospital Wastewater Purification

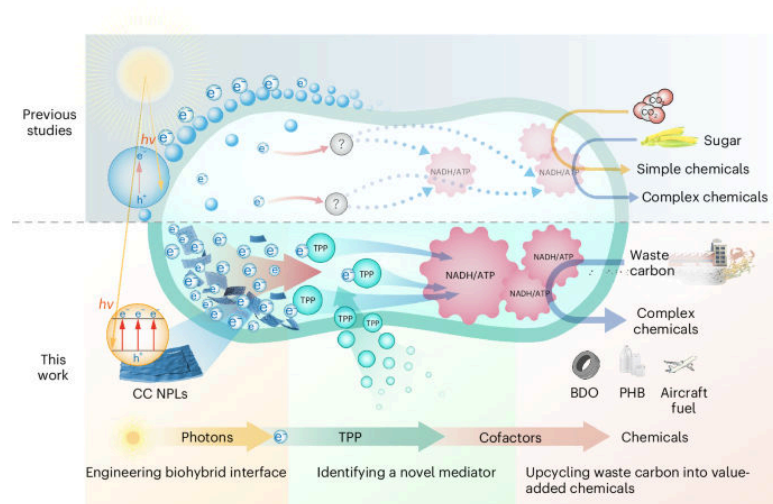
□ Environmental Science & Technology | DOI: <https://doi.org/10.1021/acs.est.6c00061>

□ 摘要: Environmental Science & Technology DOI: 10.1021/acs.est.6c00061

14. Rewiring energy flow in biohybrids for enhanced solar-driven biosynthesis

□ Nature Sustainability | DOI: <https://doi.org/10.1038/s41893-026-01787-x>

□ 摘要: Nature Sustainability, Published online: 10 March 2026; doi:10.1038/s41893-026-01787-xSolar-driven biosynthesis using semiconductor biohybrids has the potential to achieve sustainable chemical production, but often it is hindered by inefficient solar energy conversion. Here the authors develop design strategies to tackle this technical challenge.



15. [ASAP] Beyond Microplastics: How Tire Wear Particles Influence Plant Performance

□ Environmental Science & Technology | DOI: <https://doi.org/10.1021/acs.est.5c16398>

□ 摘要: Environmental Science & TechnologyDOI: 10.1021/acs.est.5c16398

16. [ASAP] Granular Stratification Enables Stable and Efficient Nitrogen Removal in Anammox Sludge Bed Reactors

□ Environmental Science & Technology | DOI: <https://doi.org/10.1021/acs.est.5c17171>

□ 摘要: Environmental Science & TechnologyDOI: 10.1021/acs.est.5c17171

17. Innovative methodologies for elucidating bushfire smoke-induced pathophysiological mechanism

□ Science of the Total Environment | DOI: <https://www.sciencedirect.com/science/article/pii/S0048969726003062>

□ 摘要: Publication date: 15 April 2026Source: Science of The Total Environment, Volume 1025Author(s): Keshav R. Paudel, Karl J. Hegarty, Jesus Shrestha, Kurtis F. Budden, Nikhil T. Awatade, Tayyaba Sadaf, Vamshikrishna Malyala, Shafagh Waters, Paris C. Papagianis, Jane E. Bourke, Peter A. Wark, Philip M. Hansbro

18. Microbial dispersal from a hyperactive sandsheet in the Icelandic Highland

□ Science of the Total Environment | DOI: <https://www.sciencedirect.com/science/article/pii/S0048969726003207>

□ 摘要: Publication date: 15 April 2026Source: Science of The Total Environment, Volume 1025Author(s): Nathan Hadland, Christopher W. Hamilton, Peter Schroedl, Federica Calabrese, Jeffery Marlow, Solange Duhamel

19. In situ degradation of biodegradable bio-based plastics in urban soil: Pilot study for PLA, PHB, PHBH, and Bio-PBS in central Tokyo, Japan

□ Science of the Total Environment | DOI: <https://www.sciencedirect.com/science/article/pii/S0048969726002937>

□ 摘要: Publication date: 15 April 2026Source: Science of The Total Environment, Volume 1025Author(s): Ayano Kobayashi, Yoshiki Harada, Yoshiharu Mitoma

20. Potential One Health risk of downwind exposure to airborne pathogens from thawing Siberian permafrost

☐ Environmental Research Letters | DOI: <https://iopscience.iop.org/article/10.1088/1748-9326/ae4819>

☐ 摘要: Permafrost degradation in Arctic and Subarctic regions, accelerating due to rapid climate change, exposes vast extents of ice- and organic matter-rich soils directly to the atmosphere. This ancient permafrost is known to harbor metabolically active microbes, including potentially harmful pathogens that have survived for thousands of years under extreme environmental conditions. The release and subsequent long-range dispersal of these organisms via atmospheric transport presents a significant, ye...

21. Generation of inner core anisotropy by anisotropic thermal conductivity of iron crystals

☐ Nature Geoscience | DOI: <https://doi.org/10.1038/s41561-026-01916-3>

☐ 摘要: Nature Geoscience, Published online: 05 March 2026; doi:10.1038/s41561-026-01916-3 Seismic wave velocity variations, or anisotropy, in the Earth's inner core may be generated by the differing thermal conductivity of iron crystals along their long and short crystallographic axes, according to coupled thermo-mechanical modelling.

22. Mantle oxidation influenced by reduction-oxidation budget of Mariana-type subduction zones

☐ Nature Geoscience | DOI: <https://doi.org/10.1038/s41561-026-01939-w>

☐ 摘要: Nature Geoscience, Published online: 04 March 2026; doi:10.1038/s41561-026-01939-w Mariana-type subduction redox is controlled by sulfide oxidation that enables fluids to carry redox budgets from the slab to the mantle wedge, and iron-rich slab sediment melts that oxidize the back-arc mantle, according to a 2D thermomechanical-thermodynamic model.

23. [ASAP] Colonization of Microplastics by Different Strains of *Pseudomonas Syringae* Increases Ice-Nucleation Activity

☐ Environmental Science & Technology | DOI: <https://doi.org/10.1021/acs.est.5c18769>

☐ 摘要: Environmental Science & Technology DOI: 10.1021/acs.est.5c18769

24. [ASAP] Atomic Building Blocks of Global Buildings

☐ Environmental Science & Technology | DOI: <https://doi.org/10.1021/acs.est.5c11079>

☐ 摘要: Environmental Science & Technology DOI: 10.1021/acs.est.5c11079

25. A laboratory study of chemicals in nail products and exposure assessment using USEtox

☐ Science of the Total Environment | DOI: <https://www.sciencedirect.com/science/article/pii/S0048969726003025>

☐ 摘要: Publication date: 15 April 2026 Source: Science of The Total Environment, Volume 1025 Author(s): Lei Huang, Tiglath Moradkhan, Efrem Neuwirth, Jennifer Branyan, Abigail Noble, Wayne Halozan, Lan Wang, Sayaka Takaku-Pugh, Christine Papagni, Anne-Cooper Doherty

26. Hydrologic variability drives environmental and geospatial relationships in Smallmouth Basse (*Micropterus dolomieu*) distribution

☐ Science of the Total Environment | DOI: <https://www.sciencedirect.com/science/article/pii/S0048969726002226>

☐ 摘要: Publication date: 15 April 2026 Source: Science of The Total Environment, Volume 1025 Author(s): Sarah F. Sorensen, J. Tyler Fox, Daniel D. Magoulick

27. Mapping the supply of nature's contributions to people on Mount Kilimanjaro

☐ Environmental Research Letters | DOI: <https://iopscience.iop.org/article/10.1088/1748-9326/ae45bd>

☐ 摘要: Mount Kilimanjaro, with its steep elevational gradient (770 – 5886 m a.s.l.) and pronounced land-use heterogeneity, supports high biodiversity and diverse nature's contributions to people (NCP), but it is underrepresented in global spatial assessments. We address this gap by mapping NCP supply across 12 ecosystem types on the southern slopes identifying hotspots and coldspots and quantifying synergies and trade-offs among NCP categories. We use 25 context-specific NCP categories that integrate loc...

28. The effects of self-governance arrangements on the climate change resilience of small-scale fisheries in Mexico

☐ Global Environmental Change | DOI: <https://www.sciencedirect.com/science/article/pii/S095937802600021X>

☐ 摘要: Publication date: July 2026Source: Global Environmental Change, Volume 98Author(s): Xochitl Édúa Elías Ilosvay, Sarah Elkin, Sebastian C.A. Ferse, Eréndira Aceves-Bueno, Javier Tovar-Ávila, Jhosafat Rentería-Bravo, Irving Alexis Medina Santiago, Jorge García Molinos, Elena Ojea

29. Stakeholder engagement in climate change mitigation: A focus on Tanzania's renewable energy sector

☐ Climatic Change | DOI: <https://link.springer.com/article/10.1007/s10584-026-04132-5>

30. Climate change anxiety in daily life: prevalence and distribution among young people in Canada

☐ Climatic Change | DOI: <https://link.springer.com/article/10.1007/s10584-026-04141-4>

31. Sustained weakening in the Atlantic Niño/Niña variability post 2100

☐ Environmental Research Letters | DOI: <https://iopscience.iop.org/article/10.1088/1748-9326/ae49a1>

☐ 摘要: The Atlantic Niño/Niña, akin to the Pacific El Niño-Southern Oscillation (ENSO) and the Indian Ocean Dipole (IOD), is a dominant mode of interannual climate variability, exerting profound impacts on the global climate. Previous studies show reversed changes in ENSO and IOD variability before and after 2100. However, the projected evolution of the Atlantic Niño/Niña beyond 2100 remains unknown. Here, we find that the Atlantic Niño/Niña variability displays a sustained weakening toward the 23rd ce...

32. Modulation of carbendazim toxicity by enhanced acanthocephalan infection in a freshwater fish

☐ Environmental Pollution | DOI: <https://www.sciencedirect.com/science/article/pii/S0269749126002915>

☐ 摘要: Publication date: 15 May 2026Source: Environmental Pollution, Volume 397Author(s): Léa Lorrain-Soligon, Constance Warnier de Wailly, Alexandre Bauer, Thierry Rigaud, Jean-Emmanuel Rollin, Simon Agostini, Beatriz Decencièrre, Alexis Millot, Aurélie Goutte

33. In situ evaluation of an active-passive sampling (APS) technique for monitoring psychoactive compounds in effluent wastewater

☐ Environmental Pollution | DOI: <https://www.sciencedirect.com/science/article/pii/S0269749126003003>

☐ 摘要: Publication date: 15 May 2026Source: Environmental Pollution, Volume 397Author(s): Allen Jun Anies, Maria Laimou-Geraniou, Maarten Quireyns, Tim Boogaerts, Cara Byns, Taja Verovšek, David Heath, Ester Heath, Alexander L.N. van Nuijs, Adrian Covaci, Ronny Blust, Raewyn M. Town

34. Amplified response of Amazon floodplain flows to rising river levels

☐ Environmental Research Letters | DOI: <https://iopscience.iop.org/article/10.1088/1748-9326/ae45be>

□ 摘要: The dynamics of Amazon floodplains is tightly connected to the adjacent river flood pulse, yet its hydrological functioning differs from that of the main channel. While the river transports water downstream, floodplain flows promote lateral connection, facilitating biogeochemical exchanges with the river and sustaining local ecosystems and riverine communities. Extreme floods in the Amazon Basin have become more frequent in recent years, and their impact on floodplain flow remains unknown. Here,...

35. Integrating target, nontarget analysis with machine learning to illuminate PFAS characteristics and health risks in Chinese cosmetics

□ Environmental Pollution | DOI: <https://www.sciencedirect.com/science/article/pii/S0269749126003039>

□ 摘要: Publication date: 15 May 2026 Source: Environmental Pollution, Volume 397 Author(s): Li Zhao, Jian Chen, Yuxin Sun, Yangjie Li, Huiru Li, Qunyue Wu

36. A global perspective: quantifying disturbance and reclamation of surface coal mining through remote sensing innovations

□ Global Environmental Change | DOI: <https://www.sciencedirect.com/science/article/pii/S0959378026000270>

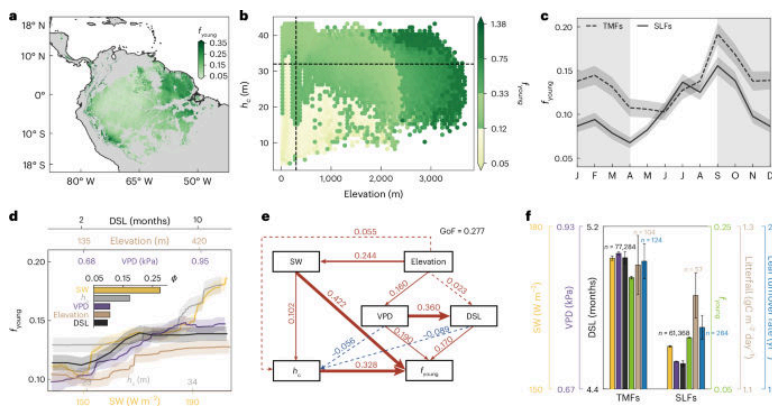
□ 摘要: Publication date: July 2026 Source: Global Environmental Change, Volume 98 Author(s): Zhen Yang, Wenrui Tian, Tingting He, He Ren, Xingdong Wang, Huawei Jiang, Like Zhao, Yanchuang Zhao, Yuwei Chen

【植物与农业】

1. Amazon rainforests are rejuvenating their canopies by producing more photosynthetically efficient young leaves under climate change

□ Nature Plants | DOI: <https://doi.org/10.1038/s41477-026-02240-9>

□ 摘要: Nature Plants, Published online: 09 March 2026; doi:10.1038/s41477-026-02240-9 The authors mapped the continental-scale fraction of age-dependent leaf area index and revealed a widespread increase in the fraction of photosynthetically efficient young leaves across the majority of Amazon rainforests over the past decades.



2. Plant Community Responses to Long-Term Nutrient Additions Interact With Elevation and Vegetation Type in a Subarctic Tundra

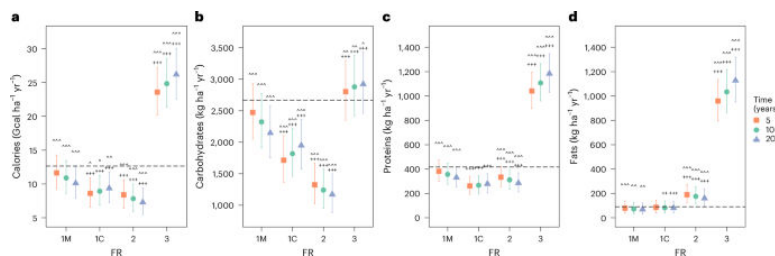
□ Journal of Vegetation Science | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70125>

□ 摘要: Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

3. Functionally rich crop rotations increase calorie and macronutrient outputs across Europe

□ Nature Food | DOI: <https://doi.org/10.1038/s43016-026-01293-5>

□ 摘要: Nature Food, Published online: 25 February 2026; doi:10.1038/s43016-026-01293-5 While the benefits of crop diversity are known, doubts remain as to whether replacing cereals in rotations reduces nutrient production. A comparison of 16 long-term field experiments across Europe shows no trade-off between functionally rich rotations and food productivity.



4. Implementation of practices shapes the effectiveness of agricultural diversification for arthropod related ecosystem services: a meta-analysis

□ Agronomy for Sustainable Development | DOI: <https://link.springer.com/article/10.1007/s13593-025-01082-7>

□ 摘要: Agricultural intensification has increased food production but has also caused significant environmental degradation and biodiversity loss. Diversification practices can mitigate these impacts while sustaining yields. While previous meta-analyses have examined their effects on arthropod populations and associated ecosystem services, focusing on individual practices or treating them as a whole, this is the first study to compare a wide range of diversification strategies across specific arthropod...



5. Effects of Ungulate Herbivores on Temperate Forest Understory Vegetation—Implications From a Large-Scale Wildlife Exclosure Experiment in Central Europe

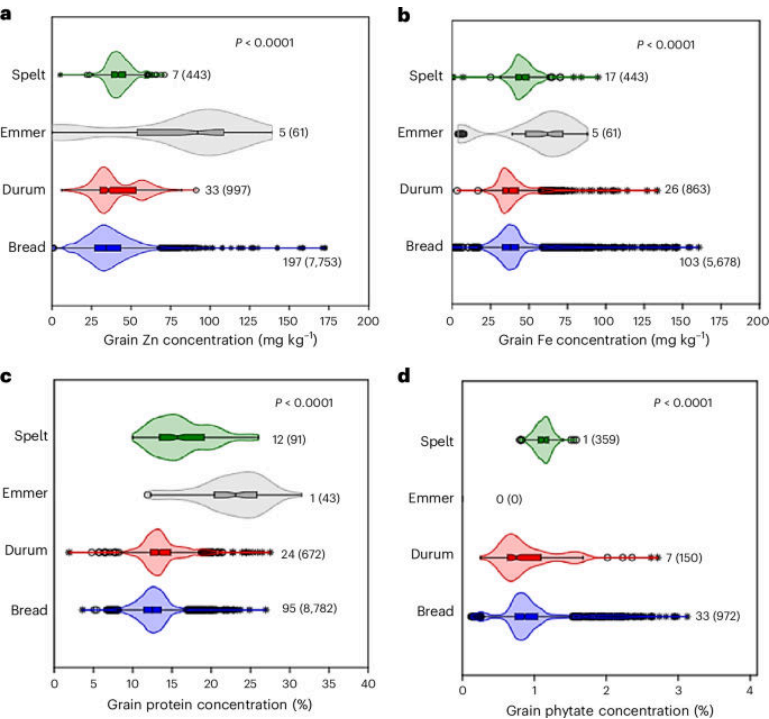
□ Journal of Vegetation Science | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70128>

□ 摘要: Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

6. Grain zinc, iron and protein concentrations of contemporary wheat cultivars fall short of targets for human health

□ Nature Food | DOI: <https://doi.org/10.1038/s43016-026-01314-3>

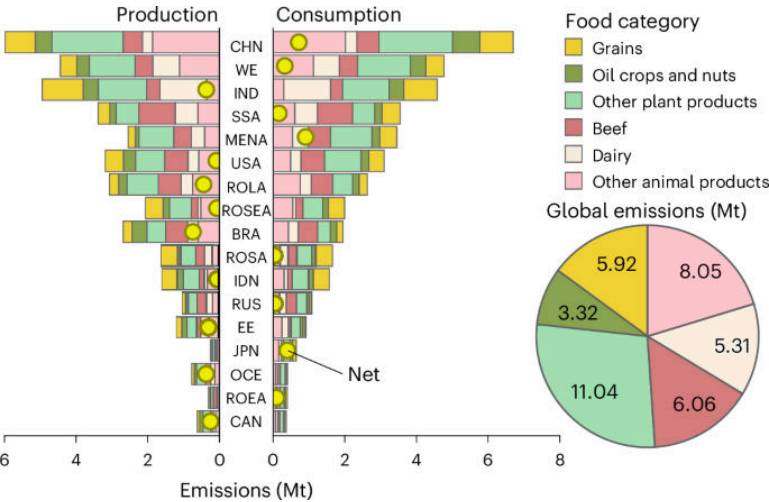
□ 摘要: Nature Food, Published online: 09 March 2026; doi:10.1038/s43016-026-01314-3A synthesis of 243 field studies shows that grain zinc, iron and protein concentrations in modern wheat cultivars have declined since the 1960s, with most varieties falling short of recommended nutritional targets. While agronomic and genetic biofortification can improve these traits, the likelihood of reaching target nutrient levels remains low, highlighting the need for integrated strategies.



7. Global food trade can mitigate substantial health burdens attributed to ambient PM_{2.5} pollution

□ Nature Food | DOI: <https://doi.org/10.1038/s43016-026-01303-6>

□ 摘要: Nature Food, Published online: 27 February 2026; doi:10.1038/s43016-026-01303-6Most estimates of air pollutant emissions and associated health burdens from global food systems focus on local food production. By integrating four advanced global models, this study estimates the impacts of food trade and consumption across distant regions.



8. Grass inflorescence morphodynamics guides yield improvement in wheat

☐ Nature Plants | DOI: <https://doi.org/10.1038/s41477-026-02246-3>

☐ 摘要: Nature Plants, Published online: 11 March 2026; doi:10.1038/s41477-026-02246-3 Morphodynamic modelling reveals how timing and fate shape diverse grass inflorescences, including wheat supernumerary spikelets. Using this model, the authors identify an early-heading paired-spikelet mutant, which exhibits robust yield gains.

9. An NLR–transposase fusion gene from rye provides broadly effective resistance to stripe rust in wheat

☐ Nature Plants | DOI: <https://doi.org/10.1038/s41477-026-02248-1>

☐ 摘要: Nature Plants, Published online: 06 March 2026; doi:10.1038/s41477-026-02248-1 Researchers have cloned the broad-spectrum stripe rust resistance gene Yr83 from rye via triticale. This gene encodes a unique NLR–transposase fusion protein, offering a pivotal genetic resource for wheat breeding.

10. Differentiation of the Seed Regeneration Niche Along a Small-Scale Plant Zonation in Mediterranean Temporary Ponds

☐ Journal of Vegetation Science | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70122>

☐ 摘要: Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

11. Neighbourly Dispute at the Edge of Life: Species Interactions Among Antarctic Mosses

☐ Journal of Vegetation Science | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70123>

☐ 摘要: Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

12. A framework for estimating manure nitrogen balance and recycling potential for current and future conditions in the USA

☐ Nature Food | DOI: <https://doi.org/10.1038/s43016-026-01312-5>

☐ 摘要: Nature Food, Published online: 09 March 2026; doi:10.1038/s43016-026-01312-5 An assessment framework is developed and applied to the USA, quantifying low rates of current manure recycling and showing that improved manure management and technological development could supply crops with nitrogen equivalent to as much as a quarter of current synthetic N fertilizer use.

13. Cross-kingdom RNA interference promotes arbuscular mycorrhiza development

☐ Nature Plants | DOI: <https://doi.org/10.1038/s41477-026-02247-2>

☐ 摘要: Nature Plants, Published online: 11 March 2026; doi:10.1038/s41477-026-02247-2 The authors provide evidence that symbiotic fungi forming widespread arbuscular mycorrhiza symbioses use cross-kingdom RNA interference to silence plant genes and promote their colonization of host roots.

14. Resurrecting the American chestnut

☐ Nature Plants | DOI: <https://doi.org/10.1038/s41477-026-02261-4>

☐ 摘要: Nature Plants, Published online: 09 March 2026; doi:10.1038/s41477-026-02261-4 Resurrecting the American chestnut

15. The Apple Does Not Fall Far From the Tree and How This Affects the Abiotic Stress Tolerance in Juvenile Trees

☐ Journal of Vegetation Science | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70126>

☐ 摘要: Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

16. Localized thresholds for smarter crop risk management

☐ Nature Food | DOI: <https://doi.org/10.1038/s43016-026-01321-4>

☐ 摘要: Nature Food, Published online: 05 March 2026; doi:10.1038/s43016-026-01321-4 Agroclimatic yield loss thresholds exhibit vital geographical variation, necessitating spatially explicit assessments over fixed assumptions amid a changing climate.

17. Twelve-year organic substitution fertilization mitigates microbial nutrient limitation and enhances soil carbon pool stability in a rubber plantation

☐ Plant and Soil | DOI: <https://link.springer.com/article/10.1007/s11104-026-08349-2>

☐ 摘要: Background Maintaining the quality of the soil organic carbon (SOC) pool is crucial for mitigating fertility degradation in rubber plantations. However, the long-term efficacy of fertilization strategies, particularly organic substitution, in enhancing SOC pool stability remains inadequately assessed. Methods Using a 12-year field trial in a mature rubber plantation on Hainan Island, China, we evaluated the effects of four fertilization regimes: no fertilization (CK), conventional chemical ferti...

18. Co-designing a research agenda for UK agroforestry using a multi-actor approach

☐ Agronomy for Sustainable Development | DOI: <https://link.springer.com/article/10.1007/s13593-026-01089-8>

☐ 摘要: There is growing recognition of agroforestry's potential to help mitigate and provide resilience to the climate and biodiversity crises. Beyond its environmental benefits, agroforestry can also enhance production and profits, making it a sustainable farming solution that is scalable. Despite this, uptake within Europe is low, and many knowledge gaps remain that need to be addressed to promote adoption and optimize the management and implementation of agroforestry systems. We co-developed a resea...

19. Optimizing straw return management to suppress plant disease and promote plant growth. A meta-analysis

☐ Agronomy for Sustainable Development | DOI: <https://link.springer.com/article/10.1007/s13593-026-01094-x>

☐ 摘要: Straw return improves soil and nutrient cycling but may aggravate diseases if mismanaged. However, the comprehensive impact of different straw return methods on plant diseases remains unclear. This study aims to clarify the effects of straw return on plant diseases and explore appropriate straw return methods. This study evaluated the effect of straw return on the reduction of plant diseases by summarizing 1352 pairs of observations from 165 peer-reviewed publications. It further analyzed the ef...

20. Utilizing high-biomass Juncao to facilitate grape growth and amend soil properties via the soil-microorganism-plant synergistic interaction

☐ Plant and Soil | DOI: <https://link.springer.com/article/10.1007/s11104-026-08397-8>

☐ 摘要: Aims This study investigated the effects of returning 'Oasis No.1' Juncao straw on saline soil amelioration and grape growth promotion. Methods A field experiment was conducted using 'Shine Muscat' grapevines under rain-shelter cultivation. Four treatments with different Juncao straw application rates were established: CK (0 kg/m²), PG1 (0.5 kg/m²), PG2 (2.5 kg/m²), and PG3 (5 kg/m²). Results Juncao straw application significantly improved soil physicochemical properties. It reduced soil bulk de...

21. Rare microbial community structure and networks in association with phosphorus forms under soil phosphorus accumulation

☐ Plant and Soil | DOI: <https://link.springer.com/article/10.1007/s11104-026-08452-4>

☐ 摘要: Aims Microbes play a significant role in the biogeochemical cycling of soil phosphorus (P). However, little is known on how P accumulation affects the microbial-driving regulation of soil P forms. Methods This study

investigated soil P forms and microbial community structure in the presence or absence of P accumulation using ³¹P-NMR spectroscopy and 16S rRNA high-throughput sequencing, respectively. Results Long-term P accumulation increased available P and orthophosphate, but decreased the prop...

22. Soil microbial taxonomic traits control the biogeography of carbon use efficiency

☐ Agriculture Ecosystems & Environment | DOI: <https://www.sciencedirect.com/science/article/pii/S0167880926001283>

☐ 摘要: Publication date: 1 July 2026Source: Agriculture, Ecosystems & Environment, Volume 404Author(s): Yihe Wang, Sijin Wang, Xinran Zhang, Ziheng Peng, Shi Chen, Haibo Pan, Beibei Chen, Jiejun Qi, Chunling Liang, Jiai Liu, Chengjie Ren, Gehong Wei, Shuo Jiao

23. Atmospheric acid deposition leads to increased soil inorganic carbon loss on the Qinghai-Tibetan Plateau

☐ Agriculture Ecosystems & Environment | DOI: <https://www.sciencedirect.com/science/article/pii/S0167880926001337>

☐ 摘要: Publication date: 1 July 2026Source: Agriculture, Ecosystems & Environment, Volume 404Author(s): Jin Hu, Zelin Huang, Xiaodong Song, Yue Dong, Huiying Wen, Shuming Ding, Fei Yang, Ganlin Zhang

24. Evaluation of pesticide reduction strategies in cropping systems. A review

☐ Agronomy for Sustainable Development | DOI: <https://link.springer.com/article/10.1007/s13593-026-01086-x>

☐ 摘要: The unsustainable and excessive use of pesticides in agriculture causes multi-faceted environmental and human health problems worldwide. Numerous strategies have been proposed to reduce pesticide use; however, the knowledge regarding their performance is scattered. To move forward, it is crucial to evaluate pesticide reduction strategies and identify the most efficient ones. The objective of this work is to globally synthesize published literature on pesticide reduction strategies in order to de...

25. Two subtropical mycorrhizal tree species differ in contributions to soil carbon pools under drought

☐ Plant and Soil | DOI: <https://link.springer.com/article/10.1007/s11104-026-08465-z>

☐ 摘要: Background and Aims This study investigated the effects of drought on carbon (C) allocation underground in two evergreen mycorrhizal tree species. Methods Two-year-old *Schima superba* (AM) and *Castanopsis sclerophylla* (ECM) seedlings underwent a short-term drought with ¹³CO₂ labeling using the in-growth core method to quantify soil C input, with C content and δ¹³C analyzed by elemental analyzer–isotope ratio mass spectrometer. Results Short-term drought had no significant effects on biomass, ¹³C ...

【综合顶刊】

1. Climate change will increase forest disturbances in Europe throughout the 21st century

☐ Science | DOI: <https://doi.org/10.1126/science.adx6329>

2. Accelerated discovery of cell migration regulators using label-free deep learning–based automated tracking

☐ Science Advances | DOI: <https://doi.org/10.1126/sciadv.aea1492>

3. This fish shouldn't exist —the weird genetics of clonal vertebrates

☐ Nature | DOI: <https://doi.org/10.1038/d41586-026-00798-0>

□ 摘要: Nature, Published online: 11 March 2026; doi:10.1038/d41586-026-00798-0A study reveals how the asexual Amazon molly defies evolutionary expectations —plus, evidence of what might be powering superluminous supernovae.



4. Overlooked and overexploited: Extensive conversion of grasslands and wetlands driven by global food, feed, and bioenergy demand

□ PNAS | DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2521183123>

□ 摘要: SignificanceVast natural ecosystems are being cleared for agriculture to meet rising domestic and international demand. While forests have received significant policy and research attention, land-use pressure is increasingly shifting to grasslands and ...

5. A hierarchical theranostic nanoagent for multimodal imaging and targeted foam cell intervention in atherosclerosis

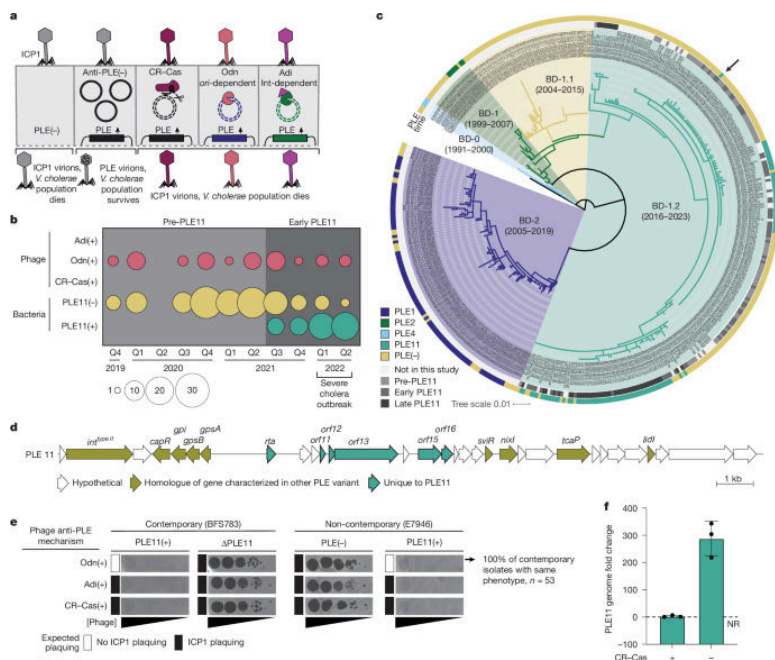
□ Nature Communications | DOI: <https://doi.org/10.1038/s41467-026-70463-7>

□ 摘要: Nature Communications, Published online: 11 March 2026; doi:10.1038/s41467-026-70463-7Foam cells play a key role in plaque destabilization in atherosclerosis, but their monitoring and regulation remain challenging. Here, the authors present a nanoagent that integrates multimodal imaging for plaque imaging and targeting. The platform allows for controlled drug release, promoting lipid efflux, reducing inflammatory signalling, and enhancing plaque stabilization in animal models.

6. Capturing dynamic phage–pathogen coevolution by clinical surveillance

□ Nature | DOI: <https://doi.org/10.1038/s41586-026-10136-z>

□ 摘要: Nature, Published online: 11 March 2026; doi:10.1038/s41586-026-10136-zThe acquisition of a parasitic anti-phage mobile genetic element, PLE11, showing potent anti-phage activity against cocirculating ICP1, and the subsequent evolution of ICP1 to escape this defense, are captured, revealing the molecular basis of the natural selection of a globally notable pathogen and its virus.



7. Profile of Emily E. Brodsky

PNAS | DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2603284123>

摘要: To study the interplay of mechanical stress and strain in faults, seismologist Emily Brodsky looks for earthquakes with a known trigger, such as a volcano or injection of underground wastewater. These triggers help explain how a fault breaks. Brodsky ...

8. Uniqueness and predictability in evolution and the history of mollusks

PNAS | DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2520986123>

摘要: Significant Unique phenomena are important in history but are difficult to study because they resist categorization and are therefore unpredictable. To examine the timing and circumstances of unique evolutionary events, we identified traits that evolved ...

9. Effects of correlated collisions and intermittency on the growth of lucky droplets

PNAS | DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2502553123>

摘要: Significant Given the complexity of precipitation in warm clouds, simple conceptual models are crucial for identifying key aspects of accelerated droplet growth. Statistical outliers with anomalously frequent collisions, so-called lucky droplets, are ...

10. Multispecies analysis of social effects on same-sex sexual behavior challenges mistaken identity hypotheses in insects

PNAS | DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2506022123>

摘要: Significant Same-sex sexual behavior (SSB) has been recorded across the animal kingdom. In insects, SSB is often thought to arise from receiver misclassification of partner sex, or “mistaken identity,” but comparative studies testing this are lacking. ...

11. Repeated convergent evolution of bradykinin mimics as defensive toxins

☐ Science | DOI: <https://doi.org/10.1126/science.adx0452>

☐ 摘要: Science, Volume 391, Issue 6789, Page 1046-1052, March 2026.

12. Nanodomain-localized formin gates symbiotic microbial entry in legume and solanaceous plants

☐ Science | DOI: <https://doi.org/10.1126/science.adx8542>

☐ 摘要: Science, Volume 391, Issue 6789, Page 1036-1045, March 2026.

13. Multiple chromosomal inversions modulate continuous local adaptation along a steep thermal cline

☐ Science | DOI: <https://doi.org/10.1126/science.ady6774>

☐ 摘要: Science, Volume 391, Issue 6789, Page 1015-1021, March 2026.

14. A persistent-range hydrogen-bonded gel polymer electrolyte enabling wide-temperature and recyclable lithium metal batteries

☐ Science Advances | DOI: <https://doi.org/10.1126/sciadv.adz1014>

15. Lifestyle change accelerates epigenetic ageing in King penguins

☐ Nature Communications | DOI: <https://doi.org/10.1038/s41467-026-70527-8>

☐ 摘要: Nature Communications, Published online: 11 March 2026; doi:10.1038/s41467-026-70527-8 For calorie-restriction-adapted King penguins, living a sedentary life at the zoo increases life expectancy; but by suppressing metabolic challenges, it also accelerates the epigenetic markers of ageing, decoupling lifespan from health.

16. NCX1 reverse mode promotes calcium-dependent Neutrophil Extracellular Trap formation and lung damage in chronic obstructive pulmonary disease

☐ Nature Communications | DOI: <https://doi.org/10.1038/s41467-026-69636-1>

☐ 摘要: Nature Communications, Published online: 11 March 2026; doi:10.1038/s41467-026-69636-1 Neutrophils are known to be important in COPD pathogenesis and calcium signaling is involved in this process. Here the authors examine the function of Na⁺/Ca²⁺ exchanger 1 (NCX1) in neutrophil function in mouse models of chronic cigarette smoke exposure and show that inhibition of NCX1 reduces neutrophil accumulation in the lungs and formation of neutrophil extracellular traps (NETs).

17. Multimodal electron microscopy of halide perovskite interfacial dynamics

☐ Nature | DOI: <https://doi.org/10.1038/s41586-026-10238-8>

☐ 摘要: Nature, Published online: 11 March 2026; doi:10.1038/s41586-026-10238-8 A multimodal in situ electron microscopy approach enables direct visualization of structural and chemical evolution in a working halide perovskite light-emitting diode with nanometre precision.

18. A novel and efficient feature extraction algorithm using kmer-derived mutation signal

☐ PeerJ | DOI: <https://peerj.com/articles/20940>

☐ 摘要: Effective algorithms for extracting genomic features are crucial for downstream bioinformatics research. Although kmer-based descriptive statistical features (frequency and position) offer a unique perspective for genomic feature extraction, their biological significance warrants further investigation. Therefore, uncovering the biological significance of kmers in genomes remains a significant challenge in current kmer-based genomic feature mining. In this work, based on reverse, complementary an...

19. Weakening vegetation control on global terrestrial evapotranspiration in a warmer world

☐ Communications Earth & Environment | DOI: <https://doi.org/10.1038/s43247-026-03372-8>

☐ 摘要: Communications Earth & Environment, Published online: 10 March 2026; doi:10.1038/s43247-026-03372-8 The positive sensitivity of terrestrial evapotranspiration to growing-season averaged leaf area index is projected to weaken over about 80% of global land area by 2100, according to integration of satellite data and CMIP6 Earth system model simulations.

20. In-situ deep ocean monitoring reveals rapid kelp degradation limits marine biomass-based carbon sequestration potential and alters benthic ecosystems

☐ Communications Earth & Environment | DOI: <https://doi.org/10.1038/s43247-026-03342-0>

☐ 摘要: Communications Earth & Environment, Published online: 11 March 2026; doi:10.1038/s43247-026-03342-0 Sinking kelp to the seafloor results in rapid biomass degradation, suggesting this technique has limited potential for carbon dioxide removal in the Northeast Pacific oxygen minimum zone, according to a year-long in-situ experiment

21. Three new *Pristimantis* species in the subgenus *Huicundomantis* (Amphibia: Anura: Strabomantidae) from Reserva Biológica Cerro Plateado, southern Ecuador

☐ PeerJ | DOI: <https://peerj.com/articles/20930>

☐ 摘要: We describe three new species of direct-developing frogs of the genus *Pristimantis*, subgenus *Huicundomantis*, from the Reserva Biológica Cerro Plateado in southern Ecuador, one of the most biodiverse yet least explored regions of the Cordillera del Cóndor. Combining molecular, morphological, and bioacoustic data, we provide diagnoses and phylogenetic placements for *Pristimantis verrucosus* sp. nov., *P. plateado* sp. nov., and *P. melanops* sp. nov. The first two species are placed within the *P. crypt.*

22. Reforestation scenarios shape global and regional temperature outcomes

☐ Communications Earth & Environment | DOI: <https://doi.org/10.1038/s43247-026-03331-3>

☐ 摘要: Communications Earth & Environment, Published online: 11 March 2026; doi:10.1038/s43247-026-03331-3 Strategic reforestation can achieve similar global cooling with half the forest area, by avoiding reforestation in high latitudes, as revealed by implementing three distinct reforestation scenarios in a fully coupled Earth System Model.

23. MADS-box gene expression patterns correlate with drought and salt stress responses in Kentucky bluegrass (*Poa pratensis* L.)

☐ PeerJ | DOI: <https://peerj.com/articles/20933>

☐ 摘要: Background Kentucky bluegrass (*Poa pratensis* L.) is a perennial cool-season turfgrass commonly utilized in lawns. However, turfgrass quality is affected by abiotic stress. Guided by previous reports demonstrating the critical involvement of MADS-box transcription factors in abiotic stress responses in monocot species such as rice (*Oryza sativa*) and sheepgrass (*Leymus chinensis*), this study specifically focuses on the MADS-box gene family to investigate their potential roles in stress adaptation me...

24. Left atrial strain reveals subclinical dysfunction in children after successful coarctation repair

☐ PLOS ONE | DOI: <https://journals.plos.org/plosone/article>

☐ 摘要: by Thuy Thuc Minh Pham, Phuc Nang Vu, Chi Thi Khanh Nguyen, Hung Quoc Nguyen, Thu-Tinh Nguyen, Phuc Minh Vu Background Although surgical repair of aortic coarctation (CoA) often restores hemodynamic integrity, subclinical myocardial dysfunction may persist long-term. Left atrial (LA) strain is a sensitive marker of

atrial compliance and early diastolic dysfunction, but its role in pediatric CoA is underexplored. Methods The study was a multicenter, cross-sectional, observational study conducted ...

【保护与生物多样性】

1. Structured machine learning modeling to support conservation of deep-sea benthic biodiversity

☐ Conservation Biology | DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70255>

☐ 摘要: Conservation Biology, EarlyView.

2. Perceived costs as drivers of wildlife management preferences in rural Tanzanian communities

☐ Conservation Biology | DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70251>

☐ 摘要: Conservation Biology, EarlyView.

3. Scenarios and strategies for future-proofing ecosystem management under climatic novelty

☐ Conservation Biology | DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70250>

☐ 摘要: Conservation Biology, EarlyView.

4. A novel kleptoplastidic symbiosis revealed in the marine centrohelid *Meringosphaera* with evidence of genetic integration

☐ Current Biology | DOI: <https://www.sciencedirect.com/science/article/pii/S0960982226001569>

☐ 摘要: Publication date: 9 March 2026Source: Current Biology, Volume 36, Issue 5Author(s): Megan E.S. Sørensen, Vasily V. Zlatogursky, Ioana Onuț-Brännström, Anne Walraven, Rachel A. Foster, Fabien Burki

5. Indigenous governance and the future of conservation

☐ Conservation Biology | DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70254>

☐ 摘要: Conservation Biology, EarlyView.

6. Integrating α and β diversity in conservation planning

☐ Conservation Biology | DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70243>

☐ 摘要: Conservation Biology, EarlyView.

7. Totipotency and high plasticity in an embryo with an invariant, fate-specifying cleavage program

☐ Current Biology | DOI: <https://www.sciencedirect.com/science/article/pii/S0960982226001636>

☐ 摘要: Publication date: Available online 10 March 2026Source: Current BiologyAuthor(s): Amber Q. Rock, Mansi Srivastava

8. Nocturnal ants navigate using a time-compensated lunar compass

☐ Current Biology | DOI: <https://www.sciencedirect.com/science/article/pii/S0960982226001600>

☐ 摘要: Publication date: Available online 10 March 2026Source: Current BiologyAuthor(s): Cody A. Freas, Ken Cheng

9. Introgression and parental conflict shape repeated occurrences of postzygotic isolation in *Mimulus*

☐ Current Biology | DOI: <https://www.sciencedirect.com/science/article/pii/S0960982226001739>

☐ 摘要: Publication date: Available online 10 March 2026Source: Current BiologyAuthor(s): Megan E. Frayer, Hagar K. Soliman, Pia F. Schwarz, Jenn M. Coughlan

10. Visual attention in peripersonal space is dependent on differential modulation of V2 feature selectivity by hand vision and proprioception

☐ Current Biology | DOI: <https://www.sciencedirect.com/science/article/pii/S0960982226001442>

☐ 摘要: Publication date: 9 March 2026Source: Current Biology, Volume 36, Issue 5Author(s): Hamidreza Ramezani, Devin Heinze Kehoe, Carolyn J. Perry, Maziar Fallah

11. Assessing the impact of linear infrastructure on landscape connectivity: a worked example from South Sudan and Uganda

☐ Biodiversity and Conservation | DOI: <https://link.springer.com/article/10.1007/s10531-026-03311-1>

☐ 摘要: Human infrastructure development is a major cause of environmental fragmentation, which threatens biodiversity globally. Fragmentation and its consequences are particularly pronounced in developing countries, where rapid economic growth and infrastructure development often have devastating impacts on biodiversity and ecosystems. Strategies to halt biodiversity loss require the protection of ecological connectivity between areas of high conservation importance, but this is often overlooked in env...

12. Persistence of medium- and large-sized mammals in an oil palm-dominated landscape of Sarawak, Malaysian Borneo

☐ Biodiversity and Conservation | DOI: <https://link.springer.com/article/10.1007/s10531-026-03304-0>

☐ 摘要: Forest mammals in Sarawak are at risk of local extinction from habitat fragmentation. Remnant forests on oil palm estates are important refugia for mammal metapopulations. Metapopulation persistence depends on species' ability to traverse the oil palm matrix. Using camera traps, we examined the occupancy of medium- to large-mammals in six forest fragments (range = 29–990 ha) and the oil palm matrix in northern Sarawak. The forest fragments collectively contained 22 of 24 mammal species occurring...

13. Computer vision-based population monitoring of immobile synchronous firefly swarms

☐ Biodiversity and Conservation | DOI: <https://link.springer.com/article/10.1007/s10531-026-03297-w>

☐ 摘要: Southeast Asia hosts a firefly species, *Pteroptyx malaccae*, that holds incredible value both scientifically and culturally. These insects exhibit synchronous swarm flashing behavior for long periods while remaining immobile. This offers a unique opportunity for scientific study: it is possible to film the fireflies and label each individual with a unique virtual ID that is preserved over the course of a video. From early March 2023 until the end of February 2024, we gathered data in the form of ...

14. Species-Specific Responses to Multiple Climatic Variables Predict Diverging Locations of Future Climate Change Refugia

☐ Diversity and Distributions | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ddi.70158>

15. Advances in assessing the conservation status of Italian marine habitats and invertebrates under the Habitats Directive

☐ Biodiversity and Conservation | DOI: <https://link.springer.com/article/10.1007/s10531-026-03296-x>

☐ 摘要: Conservation plays a crucial role in maintaining and restoring biodiversity and ecosystems integrity. Its importance is underscored by several policy frameworks, including the Habitats Directive (HD; 92/43/CEE), which identifies habitats and species of Community Interest for which a “favourable conservation status” must be ensured. The conservation status (CS) of these habitats and species is evaluated every six years at the national level, following harmonised guidelines established at the Euro...

16. A bacterial ecocline in *Klebsiella pneumoniae* may explain its backbone phylogeny

☐ PLOS Biology | DOI: <https://journals.plos.org/plosbiology/article>

☐ 摘要: by Siqi Liu, Sarah L. Svensson, Daniel Falush The genetic structure of bacterial species is most often interpreted in terms of demographic processes such as clonal descent, but can also reflect natural selection and hence give functional and ecological insight. *Klebsiella pneumoniae* (KP) disperses effectively around the world and has high recombination rates, which should result in the species having a well-mixed gene pool. Nevertheless, phylogenies based on diverse KP strains contain a “backbon...

17. Patterns in the abundance and biomass of fauna inhabiting rhodolith beds of varying structure: the potential role of facilitation cascades

☐ Biodiversity and Conservation | DOI: <https://link.springer.com/article/10.1007/s10531-026-03269-0>

☐ 摘要: Rhodolith beds are globally distributed benthic habitats that play a key role in supporting associated faunal communities and serve as important biodiversity hotspots. Their ecological importance has been attributed to their ability to increase habitat complexity. In this context, rhodoliths are considered habitat-forming species (primary facilitator) that potentially support the establishment of further habitat providers, such as epiphytic macroalgae (secondary facilitator), which contribute to...

18. Expanding invasive species impact assessments to the ecosystem level with EEICAT

☐ PLOS Biology | DOI: <https://journals.plos.org/plosbiology/article>

☐ 摘要: by Laís Carneiro, Daniel Pincheira-Donoso, Boris Leroy, Sandro Bertolino, Morelia Camacho-Cervantes, Ross N. Cuthbert, Alok Bang, Jane A. Catford, Josie South, Steven J. Cooke, Elena Angulo, Franck Courchamp The ecological impacts of biological invasions vary widely in type, scale, and severity, highlighting the need for consistent assessment tools. The Environmental Impact Classification for Alien Taxa (EICAT) provides a standardized framework for assessing their effects, but focuses mainly on ...

19. Practical and Ethical Issues in Big Data and Machine Learning Forecasts of Zambian Community Forestry Engagement

☐ Conservation Letters | DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/con4.70022>

☐ 摘要: Conservation Letters, Volume 19, Issue 2, March/April 2026.

20. Estimating wolf population size in France using non-invasive genetic sampling and spatial capture-recapture models

☐ Biological Conservation | DOI: <https://www.sciencedirect.com/science/article/pii/S0006320726000807>

☐ 摘要: Publication date: May 2026 Source: Biological Conservation, Volume 317 Author(s): Cyril Milleret, Christophe Duchamp, Sarah Bauduin, Cecile Kaerle, Agathe Pirog, Guillaume Queney, Olivier Gimenez

21. Impacts of parental age and inbreeding on fitness in a wild insect

□ PLOS Biology | DOI: <https://journals.plos.org/plosbiology/article>

□ 摘要: by Tom Tregenza, Rolando Rodríguez-Muñoz, Alfredo F. Ojanguren, Paul Hopwood, Jelle J. Boonekamp
Parental age and inbreeding have both been shown to have substantial fitness effects in laboratory experiments and in observations of wild animals. These demographic effects are likely to be strongly impacted by habitat fragmentation and warming temperatures, so understanding them is a priority. In insects and other ectotherms, some processes implicated in senescence are dependent on temperature. Ant...

22. Cis-regulatory evolution of Wnt family genes contributes to a morphological difference between silkworm species

□ PLOS Biology | DOI: <https://journals.plos.org/plosbiology/article>

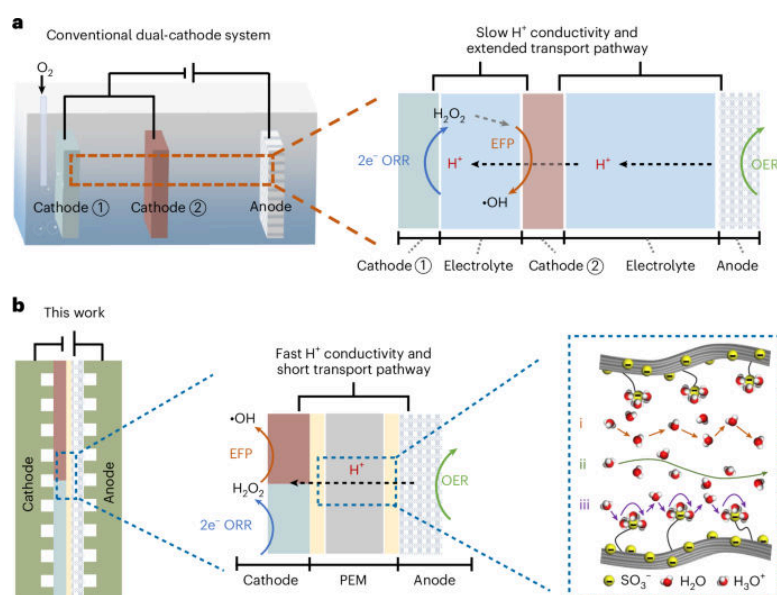
□ 摘要: by Kenta Tomihara, Ana Pinharanda, Young Mi Kwon, Andrew M. Taverner, Laura S. Kors, Matthew L. Aardema, Julia C. Holder, Lin Poyraz, Takashi Kiuchi, Peter Andolfatto
Closely related species often exhibit distinct morphologies that can contribute to species-specific adaptations and reproductive isolation. One example is Lepidopteran caterpillar appendages, such as the “caudal horn” of Bombycoidea moths, which have evolved substantial morphological diversity among species in this group. Using int...

【海洋与水生】

1. Molecular oxygen cascade reduction to •OH via coplanar dual-electrocatalytic zone achieving electrolyte-free water purification

□ Nature Water | DOI: <https://doi.org/10.1038/s44221-026-00606-z>

□ 摘要: Nature Water, Published online: 10 March 2026; doi:10.1038/s44221-026-00606-z
This study presents an electrolyte-free electrochemical system that leverages dual electrocatalytic zones to effectively remove a wide range of recalcitrant pollutants across various water matrices, including low-conductivity organic wastewater.



2. Waste per- and polyfluoroalkyl substance-assisted flash fluorination for lithium recovery from brine

□ Nature Water | DOI: <https://doi.org/10.1038/s44221-026-00593-1>

7. Meta analysis of Manning's coefficient for saltmarsh and mangroves to support broad scale assessment

☐ Estuarine Coastal and Shelf Science | DOI: <https://www.sciencedirect.com/science/article/pii/S0272771426000867>

☐ 摘要: Publication date: 1 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 334Author(s): Syed Shamsil Arefin, Robert J. Nicholls, Stefanie Nolte, Jack Heslop

8. Monitoring coastal shoreline change using PlanetScope imagery

☐ Estuarine Coastal and Shelf Science | DOI: <https://www.sciencedirect.com/science/article/pii/S0272771426000788>

☐ 摘要: Publication date: 1 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 334Author(s): Boyuan Tan, Halle Cooper, Monique LaFrance Bartley, Catherine Johnson, Sergio Fagherazzi, Cédric G. Fichot

9. A suitability evaluation method for island tourism and recreational sea use based on meta-analysis and ensemble learning

☐ Estuarine Coastal and Shelf Science | DOI: <https://www.sciencedirect.com/science/article/pii/S0272771426000909>

☐ 摘要: Publication date: 1 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 334Author(s): Cheng Zhu, Xuchun Li, Xihe Yue, Zhijie Wang, Jianli Zhang, Derui Song

10. A dose-dependent "benefit-cost" window of ammonium enrichment in eelgrass (*Zostera marina*) under non-limiting light: insights from morphology, physiology and stable isotopes

☐ Estuarine Coastal and Shelf Science | DOI: <https://www.sciencedirect.com/science/article/pii/S0272771426001046>

☐ 摘要: Publication date: 1 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 334Author(s): Shaochun Xu, Xiaoyu Li, Yi Zhou

11. When depth fails to prevent bleaching but limits coral death: insights from the 2019 heatwave in Mo'orea

☐ Coral Reefs | DOI: <https://link.springer.com/article/10.1007/s00338-026-02839-6>

☐ 摘要: Marine heatwaves (MHWs) are intensifying and increasingly threatening coral reef persistence by inducing widespread bleaching and mortality. Although depth is often proposed as a natural refuge, its protective role, especially in reducing post-bleaching coral loss, remains debated. This study quantified how depth (5, 12 and 20 m) shaped bleaching and post-bleaching mortality and evaluated drivers of depth-mediated protection during the unprecedented severe 2019 MHW in Mo'orea (French Polynesia),...

12. Evaluating satellite and modeled lake surface water temperature across the contiguous United States

☐ Hydrobiologia | DOI: <https://link.springer.com/article/10.1007/s10750-026-06178-z>

☐ 摘要: We developed a model to predict surface water temperature across U.S. lakes using satellite remote sensing and in situ observations to enhance cyanobacterial harmful algal bloom (cyanoHAB) forecasting. The study focused on Sentinel-3 Ocean and Land Colour Instrument (OLCI) sensor resolved lakes. We developed random forest models using both Landsat-derived and in-situ-measured surface water temperature. Landsat models offered broad spatial and temporal coverage of all OLCI resolved lakes, but the...

13. Cladoceran assemblage changes reveal long-term ecosystem responses to anthropogenically driven seabird declines

☐ Hydrobiologia | DOI: <https://link.springer.com/article/10.1007/s10750-026-06166-3>

☐ 摘要: While seabirds spend most of their lives in marine environments, their ecological roles extend from ocean ecosystems to terrestrial breeding sites. Here, we used subfossil cladoceran assemblages in dated sediment cores

(spanning ~ 5800 years) to track the inland aquatic impacts corresponding to changing inputs from the world's third-largest known Leach's Storm-Petrel (*Hydrobates leucorhous*) colony on Grand Colombier Island (St. Pierre and Miquelon, France), which was disrupted by European settle...

14. Flexing machine learning muscle: predicting lake calcium to prioritize invasive mussel monitoring in Alaska

☐ Hydrobiologia | DOI: <https://link.springer.com/article/10.1007/s10750-026-06127-w>

☐ 摘要: Invasive zebra (*Dreissena polymorpha*) and quagga (*Dreissena rostriformis bugensis*) mussels (ZQM) are spreading across western North America, but are not yet known to be established in Alaska. If they invaded Alaska, ZQM could disrupt freshwater ecosystems supporting some of the world's most productive wild salmon fisheries. This study examined a key factor limiting ZQM invasion risk, the concentration of dissolved ionic calcium (Ca^{2+}), by synthesizing existing water quality data from published an...

15. Main drivers of specific composition and functional trait of phytobenthos in least disturbed streams of the Pampean plain

☐ Hydrobiologia | DOI: <https://link.springer.com/article/10.1007/s10750-026-06182-3>

☐ 摘要: This study aimed to elucidate the relationships between phytobenthos and natural environmental gradients in the least disturbed stream reaches of the Pampean plain. We hypothesized that if basal conditions are maintained, variables associated with the latitudinal gradient, such as irradiance and temperature, should be the main drivers of phytobenthos and that functional trait-based approaches would be more suitable than taxonomic ones for describing these relationships. Eighteen stream reaches b...

16. Quantifying methane emissions from Laurentian Great Lakes estuaries using in situ measurements, remote sensing, and machine learning

☐ Limnology and Oceanography Letters | DOI: <https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lol2.70099>

17. Insights from multispecies coral seeding deployments on turbid nearshore reefs

☐ Coral Reefs | DOI: <https://link.springer.com/article/10.1007/s00338-026-02843-w>

☐ 摘要: Coral seeding may help sustain coral populations in a warming ocean; however, key knowledge gaps currently temper the successful application of this method across varied reef environments. This study investigated the survival and growth of three common coral species seeded on two device designs and deployed across a broad range of benthic communities in the Keppel Islands, southern inshore Great Barrier Reef. We examined the influence of algal cover and biomass, sedimentation, water flow, and te...

18. Taxonomy and Functional Traits Modulate the Environmental Niche Dimensions of Crustacean Zooplankton Communities in Submerged Aquatic Vegetation Beds

☐ Freshwater Biology | DOI: <https://onlinelibrary.wiley.com/doi/10.1111/fwb.70193>

19. The evolution, development, and genetic basis of cichlid hyoid shape reveals novel bone features and candidate genes

☐ Hydrobiologia | DOI: <https://link.springer.com/article/10.1007/s10750-026-06172-5>

☐ 摘要: Among teleosts, the hyoid bone is critical for feeding, but the causes and consequences of hyoid shape variation are poorly understood relative to other elements in the head. We leverage the experimental tractability and evolutionary richness of East African cichlids to fill this knowledge gap. We first quantified hyoid shape variation in an ecologically diverse set of species, as well as in an F5 hybrid population derived from crossing two Lake Malawi cichlid species. We found that hyoid shape ...

20. Multibeam seabed bathymetry of the Long-Term Ecological Research observatory HAUSGARTEN (Fram Strait)

☐ Deep Sea Research II | DOI: <https://www.sciencedirect.com/science/article/pii/S0967064526000305>

☐ 摘要: Publication date: June 2026Source: Deep Sea Research Part II: Topical Studies in Oceanography, Volume 227Author(s): Boris Dorschel, Simon Dreutter, Lilian Boehringer

21. Flow-aligned frameworks: linking benthic currents to cold-water coral mound development using 3D photogrammetry and ADCP data

☐ Coral Reefs | DOI: <https://link.springer.com/article/10.1007/s00338-026-02846-7>

☐ 摘要: Scleractinian cold-water corals (CWC) rely on benthic currents for the delivery of food and other resources that support their growth and subsequent mound development in the deep sea. However, the CWC-current relationship remains poorly understood, as current knowledge is predominantly derived from spatially constrained ex-situ experimentation and coarse-scale modeling. Therefore, we combined long-term in situ acoustic Doppler current profile measurements with high-resolution photogrammetric data...

22. Acoustic reaction of snapping shrimp (Alpheidae) to changing environmental variables in a coral reef system

☐ Coral Reefs | DOI: <https://link.springer.com/article/10.1007/s00338-026-02842-x>

☐ 摘要: The sounds produced by snapping shrimp (Alpheidae) are among the most common and dominant biological sources in shallow-water soundscapes worldwide and are closely linked to environmental variability, particularly in biodiverse coral reef habitats. In this study, a compact system was deployed at a coral reef station on the southeast coast of China to record the snap rate of local snapping shrimp over a three-month period, and the relationships between snap rate and temperature, salinity, turbidity...

23. Effects of daily temperature range (DTR) and thermal priming on acquired thermal resistance in *Porites* species in Moorea, French Polynesia

☐ Coral Reefs | DOI: <https://link.springer.com/article/10.1007/s00338-026-02840-z>

☐ 摘要: Thermal priming through exposure to a daily temperature range (DTR) has been proposed as a mechanism by which corals can acclimatize to elevated temperature. This study tested the effects of DTR on the response of corals from the shallow (3-m depth) backreef of Moorea, French Polynesia, to elevated temperature, and compared corals from two sites representing low-DTR (0.618 °C annually) and high-DTR (0.991 °C annually). Fragments of *Porites rus* were reciprocally transplanted between these sites f...

24. Satellite Solutions: Facing Chlorophyll-a Retrieval in Small Mountain Lakes in the Sierra Nevada, Spain

☐ Water Resources Research | DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2026WR043523>

【遥感与模型】

1. Groundwater-Derived Carbon Promotes Hypoxia and Acidification in a Large Tropical Estuary

☐ Journal of Geophysical Research - Biogeosciences | DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009360>

☐ 摘要: Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

2. Different Mechanisms Underlie Plant-Mediated Methane Transport in a Tidal Salt Marsh Versus an Impounded Brackish Wetland

☐ Journal of Geophysical Research - Biogeosciences | DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG008987>

☐ 摘要: Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

3. Declines in Conifer Forest Recovery and Forest Loss From Four Decades of Wildfire in California

☐ Journal of Geophysical Research - Biogeosciences | DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009105>

☐ 摘要: Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

4. Constraining Greenhouse Gas Cycling and Emissions in Africa's Largest Humic Lake

☐ Journal of Geophysical Research - Biogeosciences | DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009218>

☐ 摘要: Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

5. Double-Peaked Hyperspectral Features at 710 and 800 nm for Monitoring Coastal Chlorophyll- α Concentrations From Space

☐ Journal of Geophysical Research - Biogeosciences | DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009443>

☐ 摘要: Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

6. A satellite-based approach for estimating runoff and river discharge in the Pan-Arctic region from 2003 to 2022

☐ Remote Sensing of Environment | DOI: <https://www.sciencedirect.com/science/article/pii/S0034425726001239>

☐ 摘要: Publication date: 15 May 2026Source: Remote Sensing of Environment, Volume 338Author(s): Francesco Leopardi, Jacopo Dari, Carla Saltalippi, Luca Brocca, Espen Volden, Diego Fernández-Prieto, Peyman Saemian, Nico Sneeuw, Mohammad J. Tourian, Stefania Camici

7. InceptionFormer: A deep learning framework for UAV LiDAR point cloud completion to improve tree parameters estimation in dense forests

☐ Remote Sensing of Environment | DOI: <https://www.sciencedirect.com/science/article/pii/S0034425726001185>

☐ 摘要: Publication date: 15 May 2026Source: Remote Sensing of Environment, Volume 338Author(s): Binhuan Luo, Jian Yang, Shuo Shi, Ruilin Gan, Zhongliang Wu, Sihao Wang, Ao Wang, Lin Du, Wei Gong

8. Machine learning-based identification of key indicator species in coral reef fish assemblages

☐ Ecological Informatics | DOI: <https://www.sciencedirect.com/science/article/pii/S157495412600110X>

☐ 摘要: Publication date: Available online 10 March 2026Source: Ecological InformaticsAuthor(s): Eberhard Estephe Kana Djifack, Edi Prifti, Thomas Lamy, Florian Baletaud, Jean-Daniel Zucker, Norbert Tsopze, Laurent Vigliola, Eugeni Belda

9. Pulse fragmentation-induced uncertainty in forest LAI mapping using UAV LiDAR

☐ Remote Sensing of Environment | DOI: <https://www.sciencedirect.com/science/article/pii/S0034425726001112>

☐ 摘要: Publication date: 15 May 2026Source: Remote Sensing of Environment, Volume 338Author(s): Yuchen Bai, Philippe Verley, Tiangang Yin, Nicolas Lauret, Florence Forbes, Jean-Baptiste Durand, Grégoire Vincent

10. Downscaling method for red solar-induced chlorophyll fluorescence using modified kernel normalized difference vegetation index

☐ Remote Sensing of Environment | DOI: <https://www.sciencedirect.com/science/article/pii/S0034425726001215>

☐ 摘要: Publication date: 15 May 2026Source: Remote Sensing of Environment, Volume 338Author(s): Xia Jing, Jiaqi Zhao, Xinjie Liu, Bingyu Li, Yuanfang Zhang, Tianhao Zhao, Xiaopeng Liu

11. RURBAN-Map: A knowledge-guided framework for nationwide fine-scale urban-rural land use mapping in China

☐ Remote Sensing of Environment | DOI: <https://www.sciencedirect.com/science/article/pii/S0034425726001227>

☐ 摘要: Publication date: 15 May 2026Source: Remote Sensing of Environment, Volume 338Author(s): Qiqi Zhu, Dongyang Li, Longli Ran, Yiran Xu, Yong Zhang, Qingfeng Guan

12. Are remote sensing-based crop type classifications suitable for calculating a landscape heterogeneity metric? A data-fitness-for-purpose assessment

☐ Ecological Informatics | DOI: <https://www.sciencedirect.com/science/article/pii/S157495412600066X>

☐ 摘要: Publication date: May 2026Source: Ecological Informatics, Volume 95Author(s): Jannes Säurich, Marcel Schwieder, Sebastian Preidl, Florian Beyer, Markus Möller
